



PennState

College of Information
Sciences and Technology

Semester Overview and Reflection

COURSE:	IST 894
ASSIGNMENT:	ASSIGNMENT 13
SUBMITTED BY:	BRANTLEY PRICE
DUE DATE:	AUGUST 13TH, 2025
INSTRUCTOR:	DR. BARTOLACCI

Capstone Overview

Throughout the semester, I worked through a sequence of hands-on assignments. The course was designed to be the culmination of the Cybersecurity Analytics and Operations graduate program and covered a wide range of topic areas. These areas included, but were not limited to:

- Network Reconnaissance
- Vulnerability Management
- Offensive and Defensive Cyber Operations
- Cross-Site Scripting (XSS) Defense Techniques
- SQL Injection Defense Techniques
- Web App Development and Security
- Cryptography and Hashing
- Regulatory Compliance and Ethical Considerations
- Governance and Management

Each assignment was detailed enough to be helpful as a standalone exercise, but in many ways, they built on each other as the semester progressed. Using the right tools, methods, and execution techniques allowed me to take lessons learned from each exercise, which can be applied to real enterprise functions in the future.

The majority of the exercises dealt with placing yourself in either the attacker's shoes, typically using an instantiation of Kali Linux, or coming at things from the defensive side and neutralizing a threat that was already underway. There were instances of networks using legacy network protocols and passing clear-text traffic containing authentication credentials. I was able to view this through the eyes of an attacker, intercept the packets, and escalate privileges based on the credentials recovered from the

attack. In another few exercises, I put on a defensive cap and combed through a poorly developed web app. This web app was riddled with XSS and SQL injection vulnerabilities, which would allow the attacker to gain access via poorly executed input validation, allowing them to escalate their privileges to the root level.

The SCADA Security exercise was beneficial. ICS environments are well-known issues in the cybersecurity field, even catching the eye of the White House. Aging infrastructure is leaving gaping vulnerabilities in some of the most critical infrastructure across the country. In this particular exercise, I dealt with a system that was purpose-built and not done so with cybersecurity in mind. This is a real issue found in enterprise. There are legacy systems that were built years ago to serve a specific purpose, but they were never meant to be interconnected with anything else. Many legacy ICS environments look this way. These systems can be much more difficult to protect than modern systems. Using “bubble wrap” like a meticulously configured firewall/IDS/IPS, network segmentation, continuous network traffic analysis, and a Security Information and Event Management (SIEM) system, we can monitor and protect these highly vulnerable systems.

From a technical perspective, all exercises were completed, although I encountered some technical issues with a few. Through impromptu troubleshooting, I was able to identify problems that were not scripted, address them, and complete the exercises. This information was passed along to the professor and forwarded to Infosec. Hopefully, these issues can be resolved in the future, as I had a few classmates reach out to me, having issues getting through the exercises throughout the semester. These issues have the potential to break the learning environment for those who may lack knowledge of certain areas of the discipline.

Reflection

In its totality, the capstone course was quite valuable. I was somewhat worried that simply following instructions in a lab environment and summarizing what was done in each lab would get tedious and, through that tedium, become less worthwhile. I am happy to say, however, that as with most things in life, you get out what you put in. Throughout not just this semester, but the others as well, I have been able to develop skills in areas of need. I can operate a command line interface far more efficiently than I could two years ago. As someone with a background in applying and complying with various cybersecurity frameworks, this program has been invaluable. The capstone program cemented the fact that my efforts in this program have honed my skills.

As I reflect on this semester, a lot has happened. Since beginning this course, I have applied for retirement from the United States Air Force and have had a son. Ironically, while my focus has shifted from school to addressing the needs of a newborn, it reinforced the importance of what I have been doing in this capstone. Each one of the cyber ranges provided the opportunity to build upon what has been learned throughout the program. Whether it was web app security, SCADA system defense, or regulatory concerns, each of the assignments felt meaningful, and none of them felt dated or like a waste of time.

As I mentioned, I have a strong background in framework compliance. Often, that requires me to ensure a control has been implemented. As a manager, I typically do not handle the execution of these controls, but the validation of the execution. My goal for this program was to round myself out a little more. I wanted to add a more technical skillset to the managerial experience that I already have. This capstone confirmed that I have developed the skills that I set out to develop when I initially applied to the program, and I am pleased with the results.